

ROG annotation layers and tags

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1 Introduction

This document describes annotation layers and tags of the training corpus of spoken Slovenian ROG [13].

ROG corpus consists of:

ROG-SST, which includes selected GOS 2.1 transcriptions with manually assigned lemmas and morphosyntactic tags according to the MULTTEXT-East annotation scheme, as well as annotations according to the Universal Dependencies annotation scheme (i.e. part-of-speech categories, morphological features and syntactic dependencies). ROG-SST is distributed as CONLL-U format¹ (2014-2024) (.conllu files).

ROG-Art, which includes all the annotation layers from the ROG-SST plus additional annotation layers for prosodic units, disfluencies and dialogue acts. ROG-Art is distributed as EXMARaLDA format (.EXB files) [7]. In addition to the .EXB files, .EXS files and Rog-Art.coma file are added for searching through the annotated corpus in the EXMARaLDA EXAKT concordancer, .TRS files [2] are added for viewing the transcriptions without annotations in the Transcriber, and .TextGrid files [3] with additional prosodic annotations in the Praat (TeG folder) can be found in the ROG-Art folder.

This documentation provides information on annotation layers and tags for both ROG-SST and ROG-Art subset.

2 Data sources and file naming convention

Both ROG-SST and ROG-Art consist of data published in GOS 2.1 corpus [11]², which represents a merge of the:

- GOS 1 [12]³ and
- selected data from the Artur 1 corpus [10]⁴.

¹<https://universaldependencies.org/format.html>

²<http://hdl.handle.net/11356/1863>

³<http://hdl.handle.net/11356/1863>

⁴<http://hdl.handle.net/11356/1772>

Files preserve GOS 2.1 naming, but in order to enhance clarity and provide additional context, prefixes were added to the file names to make them more informative:

File Name Prefix	Description
Rog-Go1	The files correspond to the original version of the Spoken Slovenian Treebank [5], which was created by selecting data from the GOS 1 corpus based on proportional criteria. However, 22 files that were expanded and renamed using the Rog-Go2 naming convention are excluded here.
Rog-Go2	Twenty-two manually selected files containing speech samples, which were excluded from the Rog-Go1 subset, were expanded by adding additional data from the same speech events from the GOS 1 corpus.
Rog-Art	The speeches are excerpts taken from the Artur speech corpus.

3 Capitalization, punctuation, segmentation

3.1 Capitalization and punctuation

File Name Prefix	Description Of Capitalization And Punctuation in ROG	History of capitalization and punctuation in the original corpora (GOS 1, GOS 2.1 and/or ARTUR)
Rog-Go1 and Rog-Go2	All three subsets (Rog-Go1, Rog-Go2 and Rog-Art) follow identical manually checked capitalization and punctuation principles: lowercased transcriptions (except for proper nouns), and written-like punctuation, which includes both utterance-final punctuation, such as full stops and question marks, and utterance-medial punctuation, such as commas. While punctuation in Rog-Art remains identical as in the original ARTUR database, it had to be manually added to Rog-Go1 and Rog-Go2 data (see right). Transcription and punctuation principles are identical in both CONLL-U and EXB versions of the ROG data.	Rog-Go1 and Rog-Go2 transcriptions in the original GOS 1 corpus follow the same truecasing convention (capital letters only for proper names), but a more limited set of punctuation (?, !, and ...). The same principles were preserved, when extending GOS 1 to versions GOS 2.0 and GOS 2.1.
Rog-Art		Rog-Art transcriptions in the original ARTUR database adhered to standard orthographic conventions for capitalization and punctuation. This includes capitalizing both proper names and sentence-initial words, and using both sentence-medial and sentence-final punctuation. When adding ARTUR data to GOS 2.0, punctuation was removed, but then reintroduced in GOS 2.1, together with automatic conversion to truecase capitalization.

3.2 Segmentation

File Prefix	Name	Description Of Segmentation in ROG	History of segmentation in the original corpora (GOS 1, GOS 2.1 and/or ARTUR)
	Rog-Go1 and Rog-Go2	Segmentation in Rog-Go1 and Rog-Go2 has been inherited from the original GOS 1 and 2.1 corpus (see right) and remained unchanged. The sentence IDs are identical to those in GOS 1, ensuring long-term traceability and mapping between the datasets.	GOS 1 was manually segmented into utterances, broadly defined as a ‘syntactically, semantically and prosodically completed unit’. The same segmentation was also preserved in GOS 2.0 and 2.1. ^a
	Rog-Art	The segmentation of Rog-Art data in EXB format does not feature any explicit segmentation, as all words are on the same level, direct children of <tier> elements. In CONLL-U, Rog-Art has been split into utterances (written-like sentences) to align with the segmentation of Rog-Go1 and Rog-Go2 (see above). This was performed by merging the short segments from the original corpus (see right) based on punctuation marks. Specifically, consecutive segments within each utterance were combined into a single segment until a punctuation mark ‘.’, ‘?’, or ‘...’ was encountered at the end of the first segment, ensuring the segmentation better reflected syntactic structures. These merged sentence units are traceable through sentence IDs.	The manual segmentation in the original Artur database, which was also propagated to Gos 2.0 and Gos 2.1, was mostly prosody-based, i.e. introducing a segment split after each pause, and did not necessarily align with syntactic unit boundaries, such as syntactically complete phrases or sentences. ^b

^a“Izjava je osnovna enota govora, ki približno ustreza pojmu povedi v pisnem jeziku. Določimo jo tako, da je prozodično, semantično in skladenjsko zaokrožena enota. Pri zelo strnjenem govoru smo pozorni, da izjavo označimo na takem mestu, da je v signalu mogoče nedvoumno označiti mejo med izjavama, ne da bi pri tem odrezali konec prejšnje ali začetek nove izjave. Kjer zgornji vodili omogočata različne odločitve, imajo prednost krajše enote.” [1]

^b“Pomembno je, da segmenti niso predolge enote in da je tam, kjer naredimo mejo segmenta, dovolj premora v govoru, da lahko določimo mejo segmenta, ne da odrežemo del predhodnega ali del naslednjega fonema. /.../ Okvirno velja pravilo, da naj bo premor dolg vsaj 0,2 sekunde, lahko pa tudi manj v naslednjih primerih: (1) če bi bili sicer segmenti predolgi, tj. več kot okoli 10 sekund, (2) če se menjajo govorci ali (3) če bi v segment s hkratnim govorom tako zajeli tudi veliko nekratnega govora. Vedno pa mora biti med segmenti vsaj malo premora, da je meja med besedami vidna tudi na signalu. /.../ Idealno je, če lahko segment sovпада s stavki ali kratkimi povedmi oz. s semantično in skladenjsko kolikor mogoče zaključenimi, smiselnimi enotami, vendar pogosto to ni mogoče. Glavni vodili za meje med segmenti sta zato: • kratek premor v govoru in • dolžina segmenta, ki naj ne bo predolga, tj. več kot okoli 10 sekund.” [9]

4 Lemmas

Lemmas were first assigned automatically using the CLASSLA-Stanza model for lemmatisation of standard Slovenian 2.0 [8] and then manually validated/corrected. The manual corrections followed the lemmatization guidelines⁵.

5 Universal part-of-speech categories (upos)

Universal part-of-speech categories (UPOS column in ROG-SST files) follow the Universal Dependencies annotation scheme, documented in the following guidelines:

- general UD guidelines for part-of-speech (in English, online)⁶
- general UD guidelines for Slovenian (in English, online)⁷
- detailed UD guidelines for Slovenian (in Slovene, standalone document), v1.7⁸

Part-of-speech tags used:

Tag	Description
ADJ	adjective
ADP	adposition
ADV	adverb
AUX	auxiliary
CCONJ	coordinating conjunction
DET	determiner
INTJ	interjection
NOUN	noun
NUM	numeral
PART	particle
PRON	pronoun
PROPN	proper noun
PUNCT	punctuation
SCONJ	subordinating conjunction
SYM	symbol
VERB	verb
X	other

As described by Dobrovoljc [4], a semi-automatic procedure was employed to produce UD morphological annotation, by which the manually validated Multext-East (JOS) mor-

⁵<https://wiki.cjvt.si/books/05-lemmatization>

⁶<https://universaldependencies.org/u/pos/index.html>

⁷<https://universaldependencies.org/sl/>

⁸<https://wiki.cjvt.si/books/07-universal-dependencies/page/oznacevalne-smernice>

phosyntactic tags (see below) have been converted to UD parts-of-speech and feature combinations with the `jos2ud`⁹ conversion script. Mistakes and inconsistencies identified during subsequent dependency annotation and data validation have been manually corrected.

6 MULTEXT-East morphosyntactic tags (xpos)

Morphosyntactic tags in ROG-Art and ROG-SST were assigned automatically using the The CLASSLA-Stanza model for morphosyntactic annotation of standard Slovenian 2.0¹⁰ and then manually validated/corrected. The tags follow the Slovene Multext-East Specifications (v6)¹¹.

7 Universal morphological features (feats)

Universal morphological features (FEATS column in ROG-SST files) follow the Universal Dependencies annotation scheme, documented in the following guidelines:

- general UD guidelines for morphological features (in English, online)¹²
- general UD guidelines for Slovenian (in English, online)¹³
- detailed UD guidelines for Slovenian (in Slovene, standalone document), v1.7¹⁴

⁹<https://github.com/clarinsi/jos2ud>

¹⁰<http://hdl.handle.net/11356/1767>

¹¹<https://nl.ijs.si/ME/V6/msd/html/msd-sl.html>

¹²<https://universaldependencies.org/u/feat/index.html>

¹³<https://universaldependencies.org/sl/>

¹⁴<https://wiki.cjvt.si/books/07-universal-dependencies/page/oznacevalne-smernice>.

Tags for morphological features used (found as feature-value pairs, e.g. Tense=Pres):

Feature	Values	Description
Abbr	Yes	abbreviation
Animacy	Anim, Inanim	animacy
Aspect	Imp, Perf	aspect
Case	Nom, Gen, Dat, Acc, Loc, Ins	case
Definite	Ind, Def	definiteness or state
Degree	Pos, Cmp, Sup	degree
Foreign	Yes	foreign word
Gender	Masc, Fem, Neut	gender
Gender[psor]	Masc, Fem, Neut	possessor's gender
Mood	Ind, Imp, Cnd	mood
Number	Sing, Dual, Plur	number
Number[psor]	Sing, Dual, Plur	possessor's number
NumForm	Word, Digit, Roman	numeral form
NumType	Card, Ord, Mult, Sets	numeral type
Person	1, 2, 3	person
Polarity	Neg, Pos	polarity
Poss	Yes	possessive
PronType	Prs, Int, Rel, Dem, Tot, Neg, Ind	pronominal type
Reflex	Yes	reflexive
Tense	Pres, Fut	tense
Variant	Bound, Short	alternative form of word
VerbForm	Fin, Inf, Sup, Part, Conv	form of verb or deverbative

As described by Dobrovoljc [4], a semi-automatic procedure was employed to produce UD morphological annotation, by which the manually validated Multext-East (JOS) morphosyntactic tags (see below) have been converted to UD parts-of-speech and feature combinations with the jos2ud¹⁵ conversion script. Mistakes and inconsistencies identified during subsequent dependency annotation and data validation have been manually corrected.

8 Universal syntactic relations (head and deprel)

The annotation of syntactic relations between words (HEAD and DEPREL columns in ROG-SST files) follows the Universal Dependencies annotation scheme, documented in the following guidelines:

- general UD guidelines for syntactic relations (in English, online)¹⁶
- general UD guidelines for Slovenian (in English, online)¹⁷

¹⁵<https://github.com/clarinsi/jos2ud>

¹⁶<https://universaldependencies.org/u/dep/index.html>

¹⁷<https://universaldependencies.org/sl/>

- detailed UD guidelines for Slovenian (in Slovene, standalone document), v1.7¹⁸

Tags for dependency relations used (without subtypes of the type cc:preconj):

Tag	Description
acl	clausal modifier of noun
advcl	adverbial clause modifier
advmod	adverbial modifier
amod	adjectival modifier
appos	appositional modifier
aux	auxiliary verb
case	case marking preposition
cc	coordinating conjunction
ccomp	clausal complement
conj	conjunct
cop	copula verb
csubj	clausal subject
dep	unspecified dependency
det	determiner
discourse	discourse element
dislocated	dislocated element
expl	expletive
fixed	fixed multi-word expression
flat	flat multi word-expression
goeswith	disjointed token
iobj	indirect object
list	list
mark	marker (subordinating conjunction)
nmod	nominal modifier
nsbj	nominal subject
nummod	numeric modifier
obj	(direct) object
obl	oblique nominal (adjunct)
orphan	dependent of missing parent
parataxis	parataxis
punct	punctuation symbol
reparandum	overridden disfluency
root	root element
vocative	vocative
xcomp	open clausal complement

As described by Dobrovoljc [4], a semi-automatic procedure was employed to produce UD dependency annotation, involving automatically pre-parsed transcriptions with the customized Slovenian Trankit model [6], which were then manually checked by 2-3 independent annotators.

¹⁸<https://wiki.cjvt.si/books/07-universal-dependencies/page/oznacevalne-smernice>

9 Additional

In this tier corrections of transcription are annotated, e.g. when a word or filler in the transcription is missing, e.g. ‘nnn’ means that filler ‘nnn’ is missing in the defined segment.

Furthermore, lengthened phonemes in case of the lengthening are marked with sign (:) after the phoneme, e.g. ‘o(:)d’ means that phoneme ‘o’ in the word ‘od’ is lengthened.

10 Disfluency annotations

10.1 vocalDisfluency

Vocal disfluencies are annotated. These disfluencies are manifested through pronunciation or non-verbal/semi-verbal expressions. Tags are:

Tag	Description
lengthening	A specific phoneme or syllable within a word may be extended beyond its typical duration, or an entire word may be articulated more slowly and with prolongation. If such prolongation does not signal the end of a syntactic-semantic unit or does not serve as a rhetorical figure, it can be classified as disfluency. The annotation of this phenomenon largely depends on auditory perception and the annotator’s subjective judgement regarding whether the prolongation indicates that the speaker is pausing to gain time for speech formulation or not.
silentPause	A silent pause in speech may or may not indicate disfluency. Speakers may use pauses between sentences or statements for rhetorical purposes, among other reasons. The boundary between these two functions is neither clear nor easily defined. As a general rule, pauses occurring within a semantic-syntactic unit or statement are more often considered as disfluencies, while those between statements or sentences are less frequently classified as such. Additionally, in slow speech, only pauses that are notably long are marked, whereas in fast speech, even brief pauses can be perceived as disfluency.
filledPause	In the Slovenian speech, the mid-central vowel /ə/ is frequently used, and less commonly, the voiced consonants /m/ or /n/, in a manner perceived as filler sounds or referred to as gap fillers. Such sounds are consistently annotated as filled pauses without exception.
nonlinguisticSound	The speaker clears their throat, coughs, or produces a similar sound, indicating an issue with the speech apparatus.

Tag	Description
block	A speaker does not pronounce a particular word entirely fluently. There may be hesitation at the beginning, or a gap may be perceived in the pronunciation. However, the word is still produced in the correct form and the speaker does not restart the pronunciation.

10.2 verbalDisfluency

The inconsistencies in this group are verbal, manifested in the words spoken.

Tag	Description
disfluencyMarker	Disfluency marker is a discourse marker which primarily serves the role of signaling difficulties in speech production. Such frequent discourse markers in our data are: ne vem ‘I don’t know’, bi rekel ‘I would say’, se pravi ‘so to speak’, mislim ‘I mean’
disfluencyComment	Expressions or sentences in a text which are a commentary on difficulties in speech production are annotated as disfluency comment. They differ from disfluency markers in that their meaning is expressed propositionally. Examples: se opravičujem ‘I appologize’; kako se reče ‘how do you say’; kaj sem zdaj mislila reči ‘what was I about to say’.
unrepairedPronunciation	The speaker mispronounces the word, using the wrong sounds, and does not repair pronunciation. Example: zuženih (združenih) ‘nited (united)’.
unrepairedStructure	A syntactic unit, such as a sentence, contains a mismatch that is not an established pattern, not even in conversation or dialect, but rather an error or a change in the expected continuation of the sentence. Example: hkrati se pa pojavlja tudi nove bolezni ‘at the same time it appears new sicknesses’
abandoned	The speaker does not complete a started structure, nor corrects or changes it, but instead, for example, leaves the turn to the interlocutor; or the speaker is searching for a word, and the interlocutor takes the turn; or the speaker attempts to take over the turn but is unsuccessful; or the structure is deliberately left unfinished because a response from the interlocutor is already expected. Such examples are annotated as abandoned.
repetition	The speaker repeats a word or a phrase two or more times consecutively, but not for the purpose of emphasis. Repetition can be accompanied by silent or filled pause or lengthening or not. Examples: da tisti ki ki ‘so the one that that’; bolj ali manj v eem v ‘more or less in uhm in’; ko igram glasbo ko igram glasbo pozabim na vse probleme ‘when I play music when I play music I forget all the problems’
selfrepairPronunciation	Speaker re-pronounces the word before it is fully articulated. Examples: b() bo ‘w() will; da() določene ‘spi() specific’

Tag	Description
selfrepairLexicogrammar	The speaker revises their speech by fully or partially replacing, adding, and/or omitting one or more words. Examples: <i>nekam nikoli</i> ‘somewhere never’; <i>da človek ne poškoduje() da robot ne poškoduje</i> ‘that human does not ha() that robot does not harm’; <i>v tej v večini teh</i> ‘in these in most of these’
selfrepairRestart	The speaker abandons the initiated structure and restarts the sentence and/or thought. The initial structure is neither completed nor revised but is entirely abandoned, and the speaker proceeds with a new structure, often rephrasing the content in a different manner. Examples: <i>pa smešno je ker imajo recimo enkrat sva hotela iti gledat</i> ‘and it’s funny because they have for example once we wanted to go watching; <i>glih je š() letelo je mimo tega gorovja</i> ‘exactly pass() it flew right above these mountains’
selfrepairComplex	Multiple disfluencies can accumulate one after another in a segment of text, or the speaker can change the planned structure of the sentence but does not correct all parts already spoken that are structurally connected to the revision, or does not complete the repair sequence. Such and other similar self-repairs that do not fit into any of the above categories are annotated as complex disfluencies. Example: <i>to naše področje medijsko no medijsko je zelo široko področje novinarsko področje založniško bom tako rekla novinarsko zaro() založniško</i> ‘this our field of media well media is a very broad field the field of journalism the field of publishing I’ll put it this way journalist publi() publishing’.
nonspecified	A disfluency that does not fit into any of the specified types.

10.3 disfluencyStructure

Structure of the selfrepair verbal disfluency type is annotated.

Tag	Description
reparandum	The part of the text that will be modified.
interregnum	Anything in between the corrected part and the correction, including silent and filled pauses.
repair	The part of the text that replaces the corrected segment. In complex and incomplete self-corrections, there may be multiple repair segments. They are marked to collectively represent a coherent continuation of the preceding text.

11 traceability

Token IDs from the original GOS 2.1 corpus [14] are defined.

12 sentenceId

Sentence IDs from the original GOS 2.1 corpus [14] are defined.

13 prosodicUnits

The boundaries between prosodic units were annotated manually according to several prosodic parameters, including:

- pauses (the absence of speech, i.e., silence),
- pitch reset,
- intensity reset,
- changes in tempo,
- the presence of discourse markers,
- breathing patterns (partially coinciding with pauses),
- speaker turn-taking, and
- the auditory perception of the human annotator.

The values for the parameters pitch reset, intensity reset, and tempo changes were calculated on a syllable basis and served as supporting parameters for the human annotators who annotated the prosodic breaks. The key parameter was the auditory perception of the human annotator, irrespective of the calculated parameter values.

14 dialogActsPrimary

Dialogue acts are annotated. In case of multiple dialogue act functions only the primary function is annotated.

Basic unit of annotation is annotated according to the ISO 24617-2 guidelines as a “minimal stretch of communicative behaviour that has a communicative function”.

Tag	Description	Corresponding ISO 24617-2 category
informationProvidingFact	The speaker wants to make the addressee(s) aware of certain factual information. Examples: a fact, example, clarification, condition, cause, consequence, possibility, quotation, confirm, disconfirm. . .	information providing
informationProvidingOpinion	The speaker wants to make the addressee(s) aware of his/her own thoughts, opinion or evaluation. Examples: an evaluation, opinion, predict, will, belief, irony, agree, disagree, partially agree. . .	information providing
informationProvidingSentiment	The speaker wants to make the addressee(s) aware of his/her emotions. Examples: happiness, sadness, fear, anger, disgust, surprise.	information providing
informationSeeking	The speaker aims to obtain certain information from the addressee(s). Examples: different types of questions (but not rhetorical questions).	information seeking
actionDiscussion	The speaker commits himself to perform certain actions (commissive functions) or the speaker aims to make the addressee(s) perform certain actions (directive functions). Examples: promise, offer...; warning, request, demand, suggestion, advice, instruction. . .	commissive or directive
socialObligationsManagement	The speaker aims to explicitly manage interpersonal relationships. Examples: greeting, introducing people, expressing feelings towards the interlocutor - gratitude, love, appreciation, regret. . .	social obligation management

Tag	Description	Corresponding ISO 24617-2 category
metadiscourse	The speaker reveals the processes of attention/comprehension or speech production; managing the ongoing flow of interaction; referring to a previous or upcoming part of the same discourse. Examples: signalling comprehension; signalling production processes; closings and transmissions; initiations; referring backward; referring forward; rethorical questions. . .	feedback or turn-management or time-management or discourse-structuring or own- and partner-management
abandoned	The unit was abandoned and one cannot say what was its function.	none

15 TextGrid tiers

The TextGrid-files for prosodic analysis contain several tiers, including the original speech text, values of selected parameters, as well as the manual annotations of the prosodic units.

The individual tiers can be described as follows:

Tier 1 Contains the speaker ID (e.g., #Artur-N-G5082) for each speaker turn.

Tier 2 Contains the standardized transcription of the speech, imported directly from the Transcriber files.

Tier 3 Contains the word segmentation of the speech in standardized form.

Tier 4 Contains the annotation of discourse markers. If a token was manually recognized as a discourse marker, it was assigned the value “POS”.

Tier 5 Contains the conversational transcription of the speech, imported directly from the Transcriber files.

Tier 6 Contains the token ID for each individual token.

Tier 7 Contains the word segmentation of the speech in conversational form.

Tier 8 Contains the syllable segmentation of the conversational speech form.

Tier 9 Contains the phone segmentation of the conversational speech form.

Tier 10 Contains the value for the parameter pitch reset. If pitch reset occurs, the value is “POS”, if not, the value is “NEG”. The pitch reset values are calculated according to the syllable-level segmentation from Tier 8.

- Tier 11** Contains the value for the parameter intensity reset. If intensity reset occurs, the value is “POS”, if not, the value is “NEG”. The intensity reset values are calculated according to the syllable-level segmentation from Tier 8.
- Tier 12** Contains the value for the parameter speech rate reduction. If the speech rate drops, the value is “POS”, if not, the value is “NEG”. The speech rate reduction values are determined according to the syllable-level segmentation from Tier 8.
- Tier 13** Contains the value for the parameter pause. If absence of speech is detected, the value is “POS”, if not, the value is “NEG”. The pause values are determined according to the conversational word segmentation from Tier 7.
- Tier 14** Contains the value for the parameter speaker change. If a speaker turn is detected, the value is “POS”, if not, the value is “NEG”. The speaker change values are determined according to the syllable-level segmentation from Tier 8.
- Tier 15** Contains the manually annotated prosodic units. Each non-empty segment represents one prosodic unit and contains the token IDs of all the tokens comprising a given prosodic unit.

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